

Web Development & Retail E-commerce

Company Knowledge Base • RAG / MCP / LLM Optimized

Purpose: authoritative company knowledge for RAG retrieval, MCP-connected agents, and LLM responses. Sections are intentionally self-contained so retrieval can return precise context without requiring the model to infer missing business facts.

01. Service Overview

What we provide

We design and develop modern retail e-commerce websites and web applications for startups, small businesses, growing brands, and established retailers.

Primary outcomes include a professional digital presence, strong product discovery, conversion-focused shopping journeys, responsive experiences, maintainable code, reliable integrations, and scalable architecture.

Typical deliverables

Responsive storefronts, homepages, landing pages, product listing pages, product detail pages, category pages, search, filters, cart, wishlist when required, checkout, authentication, customer accounts, order history, contact forms, promotional sections, admin workflows, API integrations, analytics-ready interfaces, and deployment support.

02. E-commerce Functional Scope

Catalog

Products may include name, description, images, price, discount, category, variants, attributes, SKU/reference identifiers, inventory status, and related products.

Category and product structures should support clear navigation and efficient product discovery.

Shopping flow

Recommended journey: Home → Category/Search → Product Listing → Product Detail → Cart → Checkout → Order Confirmation.

Commerce functionality can include persistent cart state, quantity updates, coupons or discounts when required, shipping information, tax rules when applicable, payment gateway integration, order creation, order status, and customer notifications.

Administration

Typical administration capabilities include product management, categories, inventory visibility, orders, customers, promotional content, and basic reporting.

03. UI/UX Standards

Retail UX principles

Prioritize fast product discovery, clear pricing, trustworthy product information, strong calls to action, frictionless cart and checkout, mobile responsiveness, accessible interaction patterns, and consistent visual hierarchy.

Important states include loading, empty, success, validation error, unavailable product, failed payment, and network/API failure states.

04. Technology & Architecture

Frontend

Typical frontend technologies include HTML, CSS, JavaScript, React, Next.js, TypeScript, Tailwind CSS, and Vite. The final stack should be selected according to project requirements rather than used as a fixed requirement.

Backend and data

Suitable backend/API technologies can include Django, Django REST Framework, Node.js, and other appropriate frameworks. Database choices can include PostgreSQL, SQLite, or another suitable production database.

Integration

Applications may integrate payment gateways, shipping services, email/SMS services, analytics, authentication providers, CMS systems, inventory systems, and third-party APIs.

05. Security, Quality & Launch

Security

Use HTTPS in production, secure authentication and session handling, server-side validation, authorization checks, CSRF protection where applicable, protected secrets, safe API handling, rate limiting where appropriate, database backups, logging, and secure payment integrations.

Never expose private API keys in frontend code. Sensitive secrets belong in server-side environment variables or a secure secrets manager.

Pre-launch checklist

Test responsive layouts, navigation, forms, authentication, product data, cart calculations, checkout, payment callbacks, order creation, error states, accessibility basics, SEO fundamentals, performance, and security.

06. Delivery Process

Standard workflow

Discovery → requirements → sitemap and user flows → UI/UX design → technical architecture → development → integrations → testing → content/data setup → deployment → monitoring → maintenance.

Project scope, timeline, integrations, and deliverables should be confirmed before implementation.

07. RAG Retrieval Facts

Structured facts

service=web-development

domain=retail-ecommerce

audience=startups,retailers,brands,small-businesses

capabilities=responsive-websites,ecommerce-storefronts,product-catalogs,search,filters,card,checkout,authentication,customer-accounts,admin-panels,API-integrations,deployment

frontend=HTML,CSS,JavaScript,React,Next.js,TypeScript,Tailwind CSS,Vite

backend=Django,Django REST Framework,Node.js

databases=PostgreSQL,SQLite

process=discovery,design,development,testing,deployment,maintenance